Unit 2: Computer Hardware

Understanding Basic Units of the Computer System

Specific Objectives

- Understand the basic units of computer system (Anatomy of a Digital Computer)
- •Understand how the basic digital computer is organized
- Describe the purpose of basic units of computer systems.

- The **Anatomy of a Digital Computer** refers to the structure and organization of the main components that make up a computer system. These components work together to perform various tasks such as processing data, storing information, and interacting with external devices.

  Basic units of a computer system refer to the essential components that work together to perform input, processing, output, and storage functions.

Central Processing Unit (CPU)

The CPU consists of:

- Control Unit (CU): Manages and coordinates all operations in a computer.
- Arithmetic and Logic Unit (ALU): Performs all arithmetic and logical operations.
- Main Memory: Temporarily stores data and instructions for quick access.

Importance of CPU

- The 'brain' of the computer that executes instructions.
- Coordinates data flow and processing tasks.
- Examples: Intel Core i7, AMD Ryzen.

Main Memory (Primary Storage)

- Stores data and instructions currently in use.
- Example: RAM stores open files and running applications.
- Importance: Provides quick access to the CPU for efficient processing.

Peripherals

- Input Devices: Keyboards, mouse, scanners for user input.
- Output Devices: Monitors, printers, speakers for output.
- Storage Devices: Hard drives, SSDs for data storage.

Importance of Peripherals

- Enhance user interaction and data input/output capabilities.
- Examples: Mouse (input), Monitor (output), USB drive (storage).

Secondary Storage (Secondary Memory)

- Used for long-term data storage.
- Examples are the hard disk drive (HDD) and the solid-state drive (SSD).
- Importance: Allows permanent saving of data for future access.

Difference Between Primary and Secondary Memory

Feature	**Primary Memory (Main Memory)**	**Secondary Memory (Storage)**
Definition	Memory used by the CPU to store data currently in use	Memory used for long-term storage of data
Examples	RAM, Cache Memory	HDD, SSD, Pen Drive, Memory Card
Speed	Very fast	Slower compared to primary memory
Volatility	**Volatile** (data is lost when power is off)	**Non-volatile** (data remains even when power is off)
Use	Runs applications, stores active data	Stores files, programs, backups
Storage Capacity	Smaller (GBs)	Larger (GBs to TBs)
Cost	More expensive per GB	Cheaper per GB

Program and Data Storage

- Programs and data stored in memory for quick retrieval.
- Primary Storage (RAM) and Secondary Storage (Hard Drives).

Data and Control Paths (Buses)

- Data Bus: Transfers data between components.
- Address Bus: Specifies memory locations for data access.
- Control Bus: Manages control signals for operations.

Fetch-Execute Cycle

- **Function**: Describes how the CPU retrieves and executes program instructions from memory.
- **Steps**:
  - **Fetch**: The CPU retrieves an instruction from memory.
  - **Decode**: The instruction is decoded to determine what action is required.
  - **Execute**: The instruction is carried out by the ALU or other components.
  - **Store**: Any results are stored back into memory or used as needed.

Importance: This cycle is fundamental to the computer's operation, as it enables it to process instructions continuously and perform complex computations.

Example: Opening a web browser involves fetching and executing instructions.

Importance

- This cycle runs **continuously**.
- It is the basic working process of all computers.
- Enables the CPU to run programs smoothly and perform calculations.

Example

- **Opening a web browser:**
  CPU **fetches** the instruction → **decodes** it → **executes** it → **stores** results (like loading the homepage).

Organization of a Simple Computer

- Central Unit: Controls data processing and interactions.
- Memory Unit: Stores data and instructions.
- Input/Output: Enables communication between user and computer.

Thank you